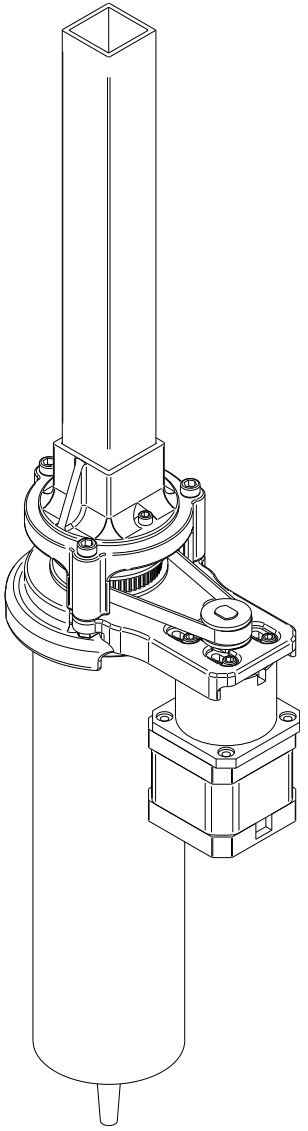
300CC Extruder V2

The 300CC V2 is the next iteration of the open-source 300CC project, designed by Emmanuel Hugnot to make paste 3D printing practical and accessible. The main goal is simple: allow anyone to experiment with materials like clay or bio-composites without needing an expensive setup. By focusing on a stripped-down design where every single part has a purpose, the entire extruder can be built for under €100.

Beyond the hardware, the project is about building a helpful community around paste printing and showing how easy it is to convert a standard FDM 3D printer into a Liquid Deposition Modeling (LDM) machine. This V2 model was developed in collaboration with third-year students at ESAD Orléans during a hands-on workshop in September 2026. Published under a Creative Commons CC BY-NC license, the design is open for makers, designers, and curious builders to use, tweak, and learn from together.

More at : emmanuelhugnot.com/300CCV2.html

CC BY NC Emmanuel Hugnot 2026



Hardware :

T8 Lead screw - 250mm + brass nut x2

6mm GT2 Belt - Perim. 200mm

Syringe 300ml/cc

Nema 17 + gearbox

Thrust bearing 21x12x5

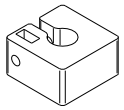
PETG Filament

15x M3-10mm
3x M4-35mm

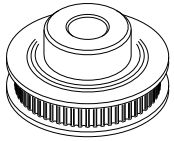
BOM link :

3D Print :

PETG
40% infill
6 walls
5 Top and Bottom
Support needed for some parts



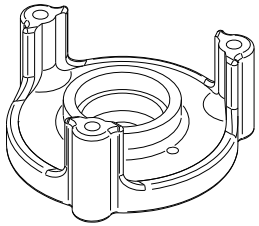
Block_tube



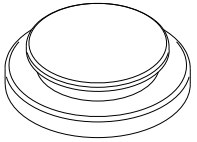
Big_pulley



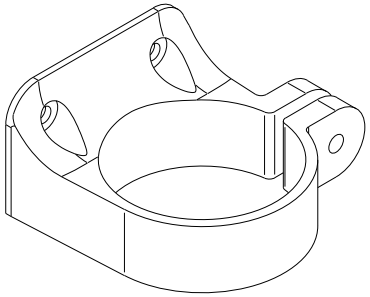
Small_pulley



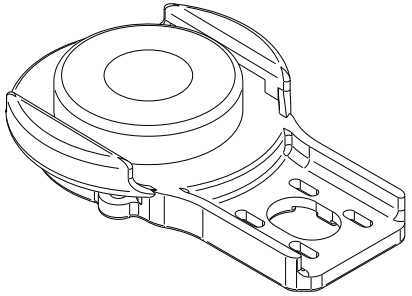
Top_mainboard



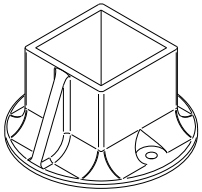
Piston



Support_extruder

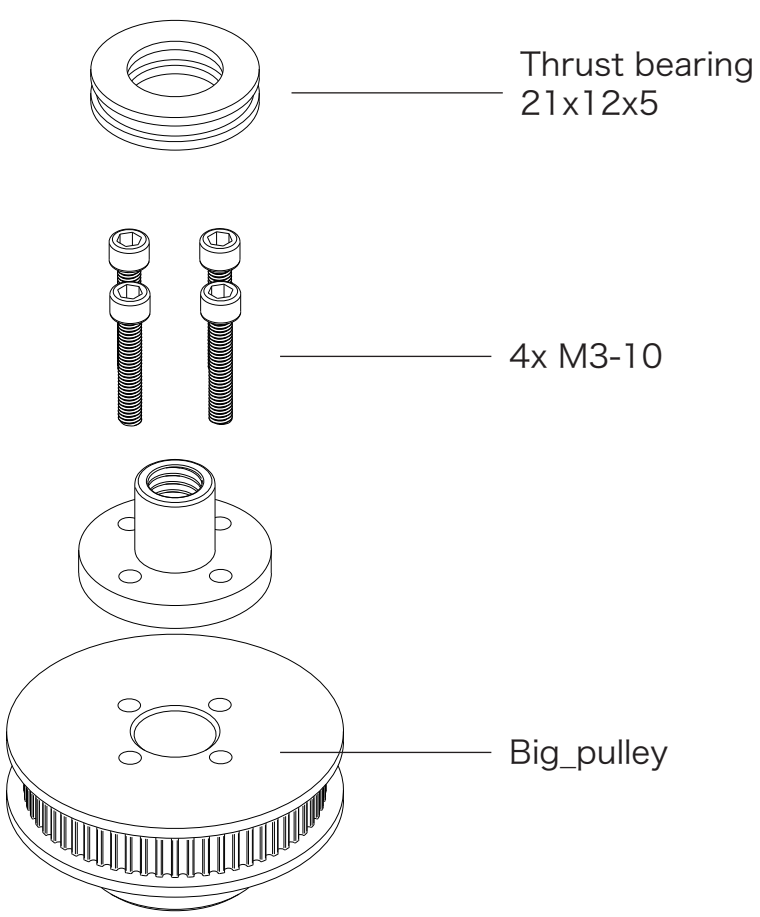


Mainboard

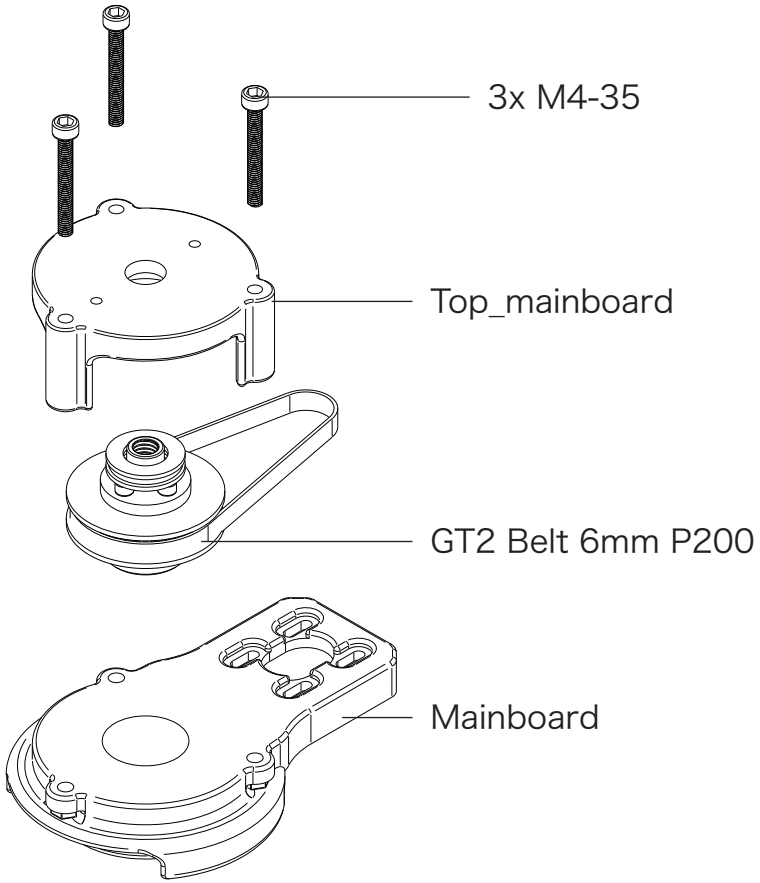


Support_tube

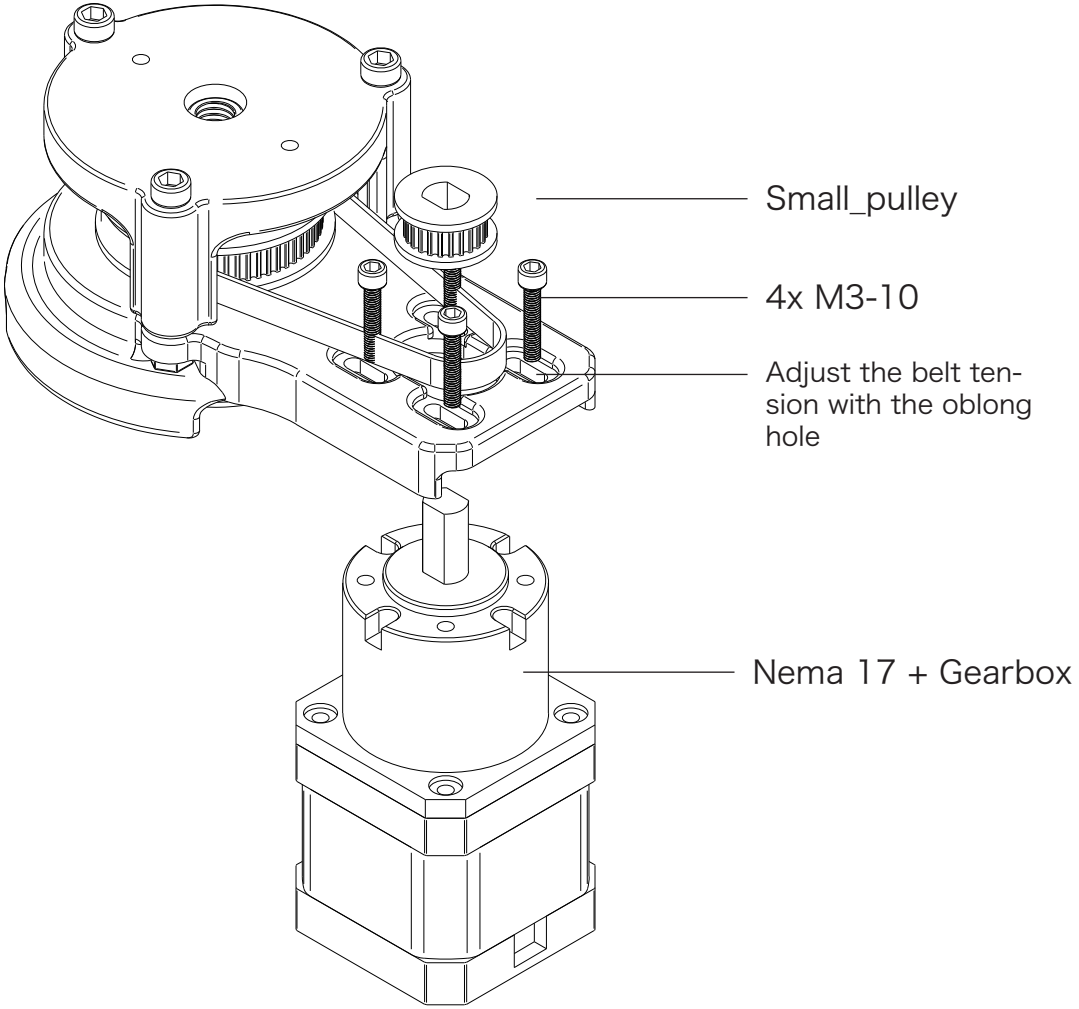
1.


Thrust bearing
21x12x5
4x M3-10
Big_pulley

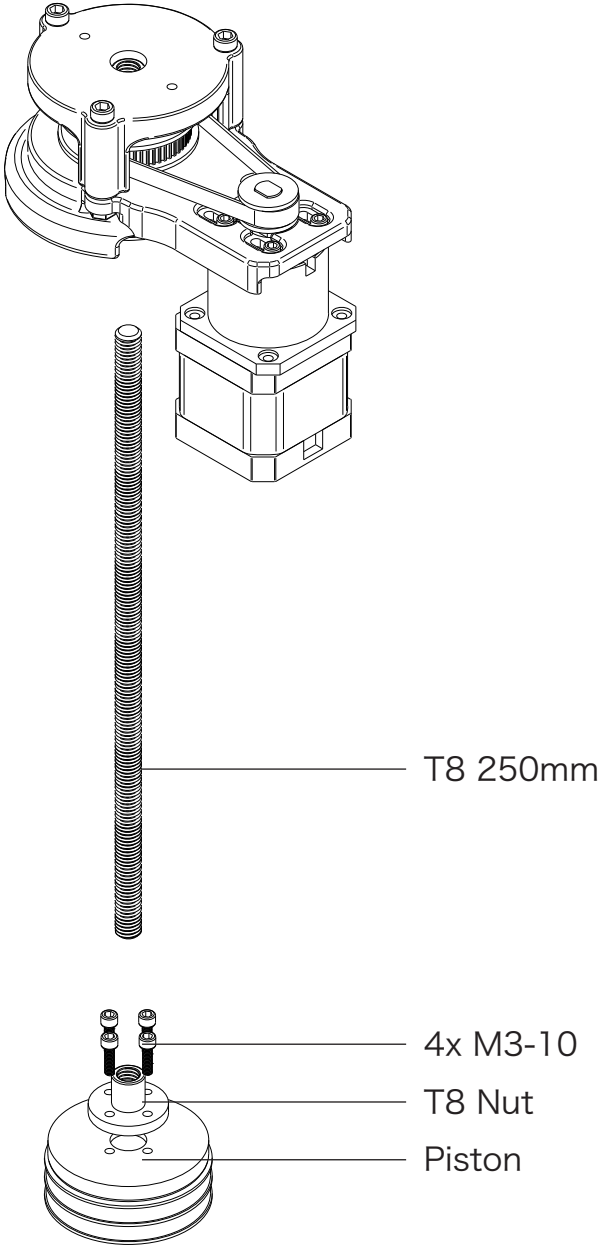
2.


3x M4-35
Top_mainboard
GT2 Belt 6mm P200
Mainboard

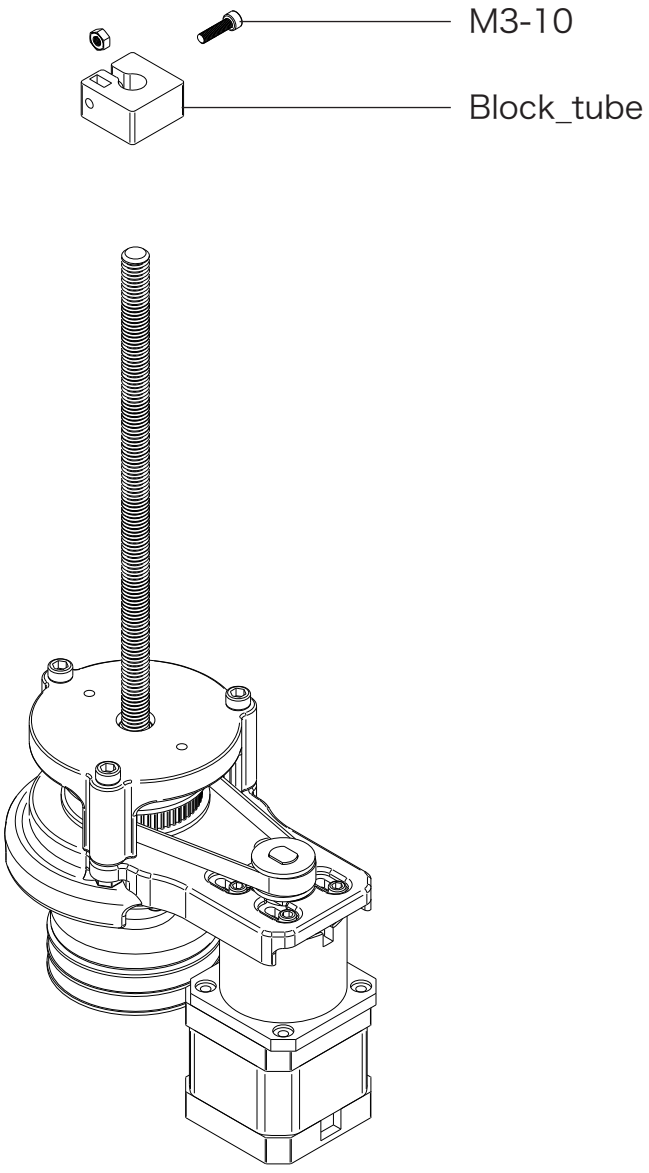
3.


Small_pulley
4x M3-10
Adjust the belt ten-
sion with the oblong
hole
Nema 17 + Gearbox

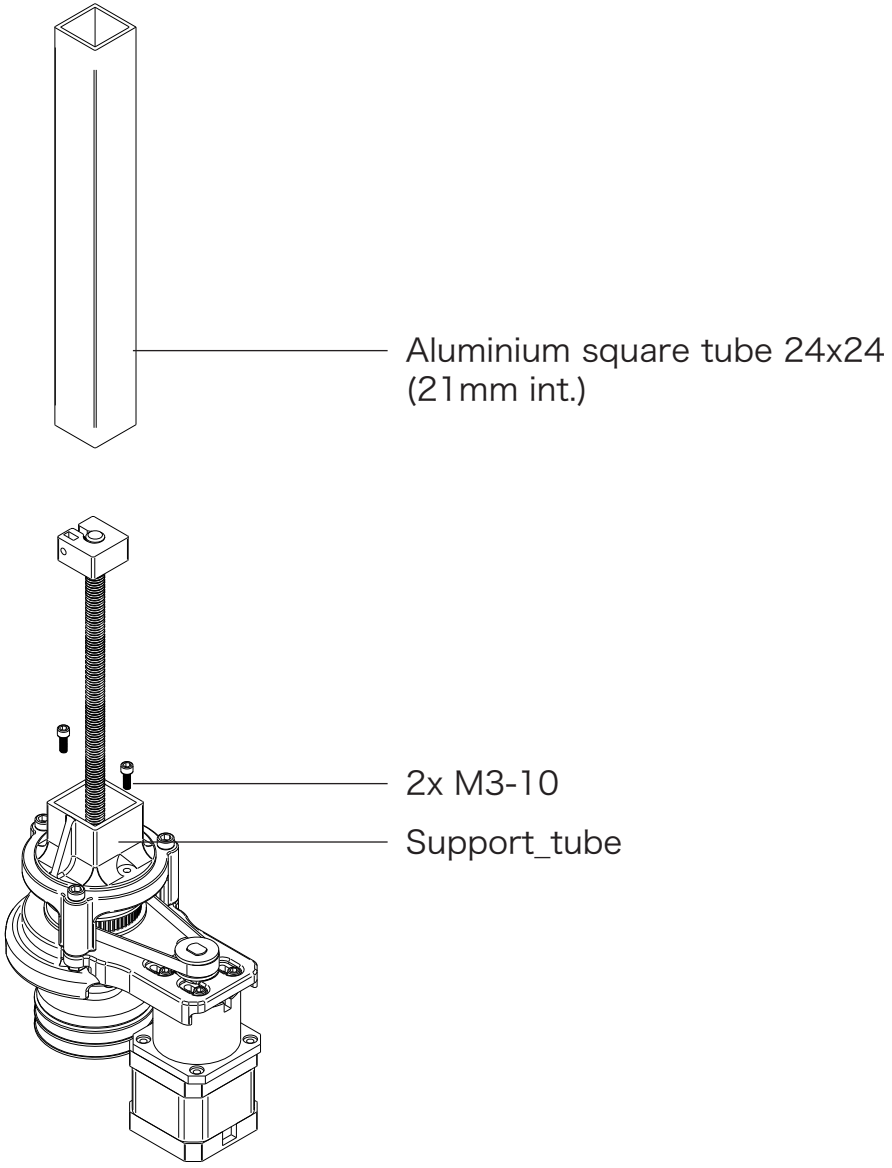
4.


T8 250mm
4x M3-10
T8 Nut
Piston

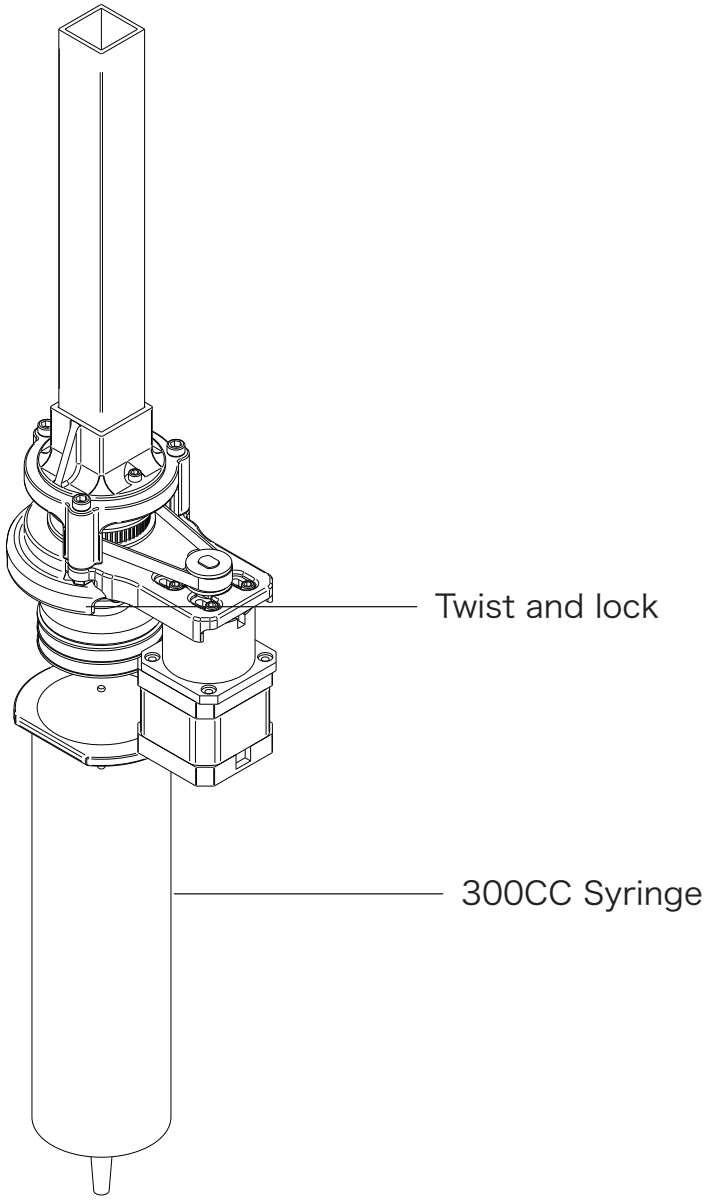
5.


M3-10
Block_tube

6.


Aluminium square tube 24x24
(21mm int.)
2x M3-10
Support_tube

7.



8.

